

Vial Wash Line Jam Detection System

Problem

The vial washline at Veolia, Boulder is a semi-autonomous process that inputs unpacked, dirty vials and feeds them through multiple sanitation stages before packaging. Due to the layout of the cleanroom this process inhabits, the line operators had to leave their workstation to inspect the rinse machine, which is the first and most important step in this process. The goal of this project was to provide a detect-and-alert system for the washline operators.

Process

I was tasked with designing a bracket for and finding an appropriate location to mount the pre-purchased Keyence IV4 vision sensor. After spending a week working with the wash line operators, I decided to mount the sensor in the rinse machine itself, in a corner location just outside the potential splash zone. I then designed and fabricated the bracket to screw directly into the framing of the rinse machine, and programmed the sensor's built-in AI tools to output OK/NG on a trigger-by-trigger basis.

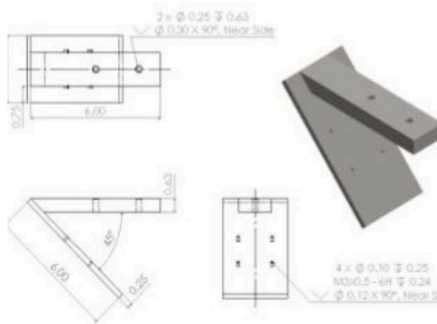


Figure 1: Bracket Dimensioned Drawing

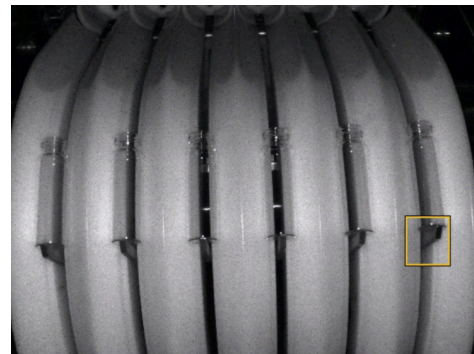


Figure 2: Sensor AI Trigger Box

Product

The final system was integrated with the wash line's PLC to provide a visual indicator at the operator's workstation. This indicator displays which lanes are occupied with vials every trigger (the OK case), and which have absent vials (the NG case), and the operators were trained to fix the vial jams if the total number of occupied lanes became less than 3. Additionally, the program detects misoriented vials and stops the rinse machine if one is detected, as such a vial will jam in the machine, causing upwards of 10 minutes of downtime each instance. This system remains operational, and to date has increased average daily OEE by 22% and decreased average daily downtime by 24%.

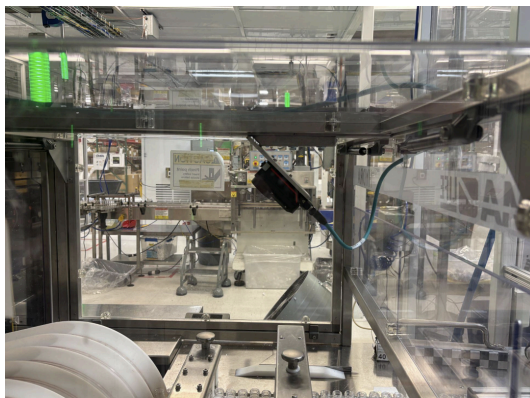


Figure 3: Installed Sensor



Figure 4: User Interface Addition*

*Image zoomed in to exclude proprietary information on operator interface